Marketing Personalization and Targeting @Uber

Ben Draves



About Me

My name is Benjamin Draves and I'm a Senior Applied Scientist at Uber in NYC. I work on building ML recommendation and targeting systems to grow Uber's consumer and earner businesses.

Data Science Topics that Excite Me

- Recommendation Systems
- Causal Machine Learning
- Graph Embeddings and Inference

Experiences & Education

- [2022-Present] Applied Scientist @ Uber
- [2017-2022] Ph.D. | M.A. in Stat @ Boston University
- [2014-2017] B.S. in Math @ Lafayette College

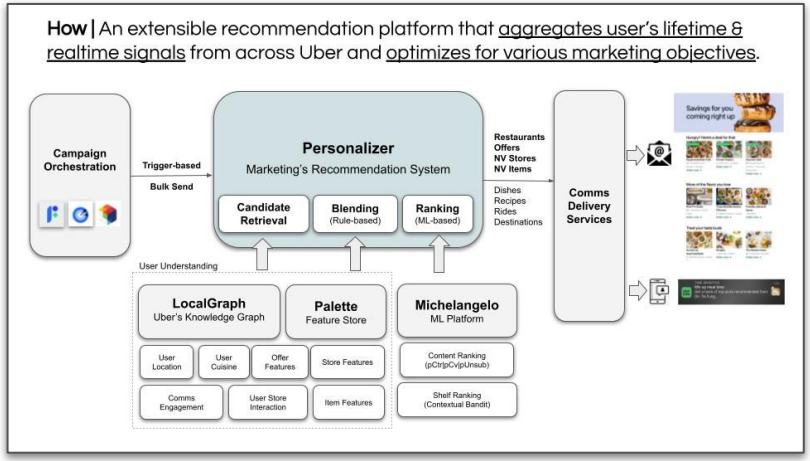
01 Data Science at Uber

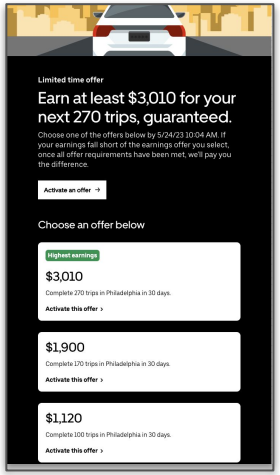
02 Data Science in Tech

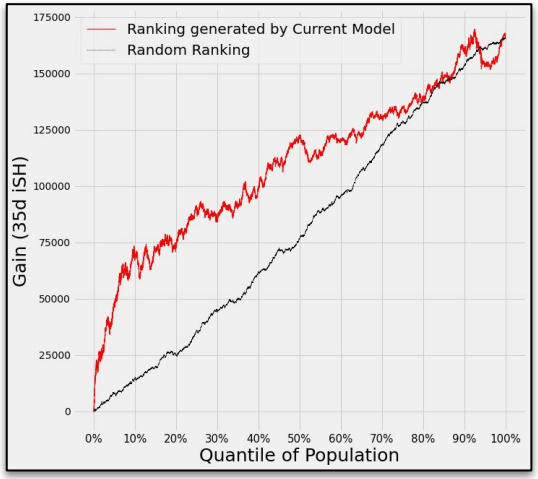
03 Discussion

Data Science @Uber

Growing Uber's Core Businesses







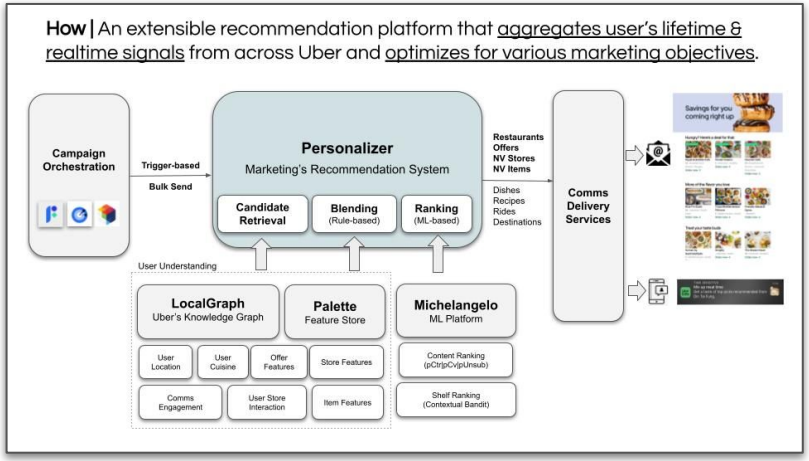
Personalization

I work on personalizing consumer's (eaters, riders, grocery and retail users, ...) experiences in out-of-app channels to drive product awareness, adoption, and growth. This oftentimes manifests in **recommending restaurants, merchants, and offers** to consumers to encourage product exploration.

Targeting

I own an incentive targeting system that identifies earners (drivers and couriers) who provide incremental supply growth to the marketplace at a reasonable cost to the business . The tech supporting this work **comes from causal machine learning** and **ads pacing systems** and

Growing Uber's Core Businesses



Personalization

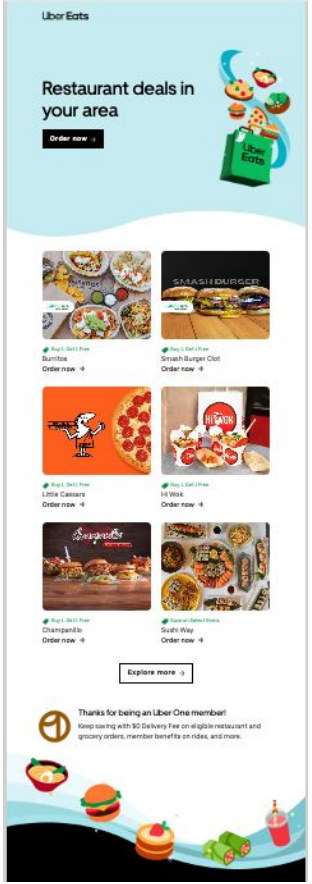
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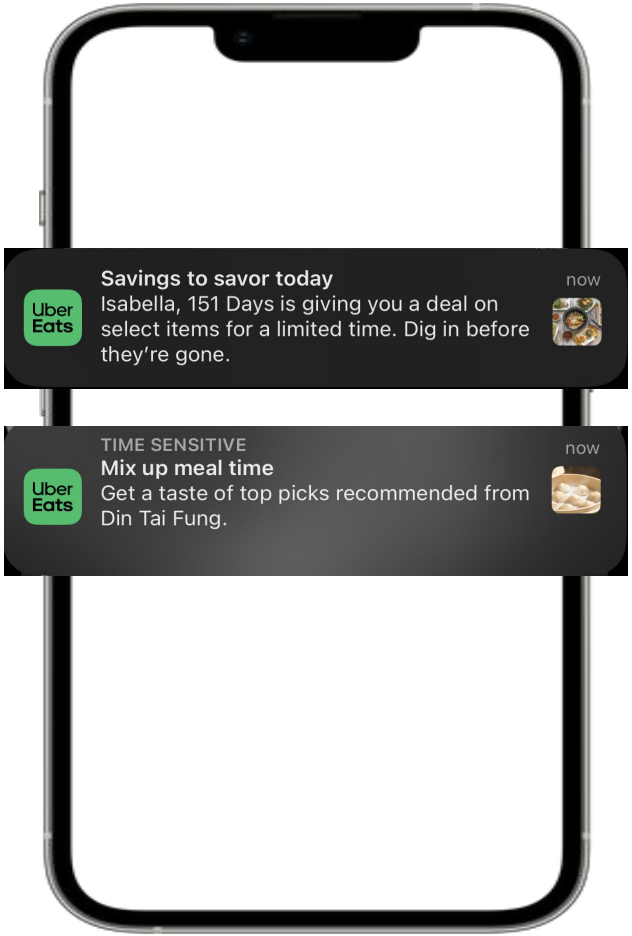
Targeting

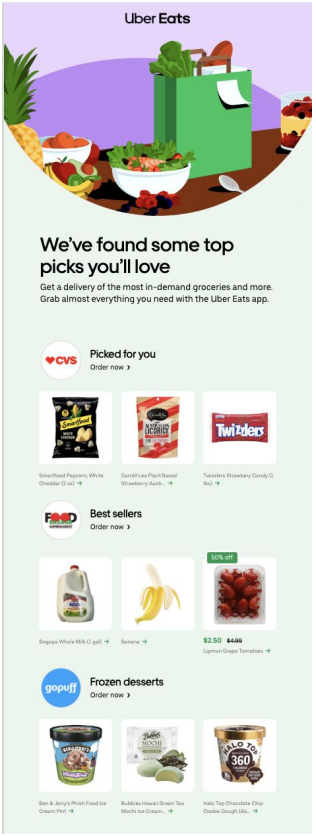
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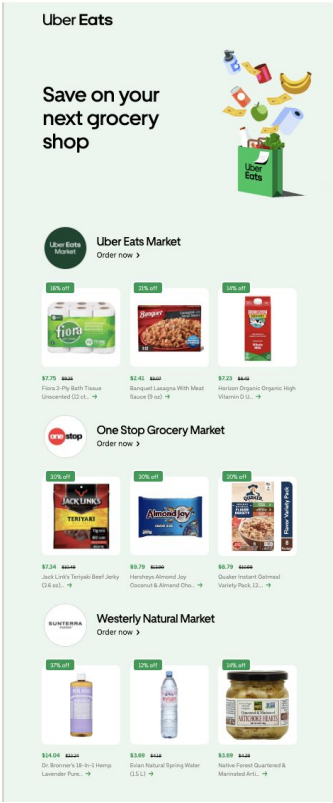
Mission | Drive growth through **personalized** out-of-app marketing communications.





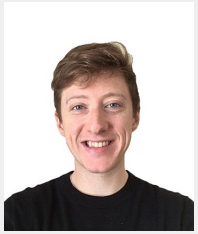






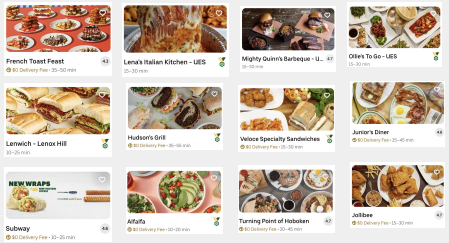
Marketing Personalizer | Personalization & Recommendation Overview

User



- User location
- Traveler behavior

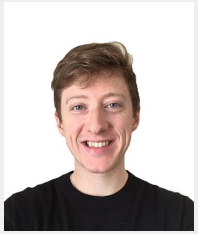
All Merchants



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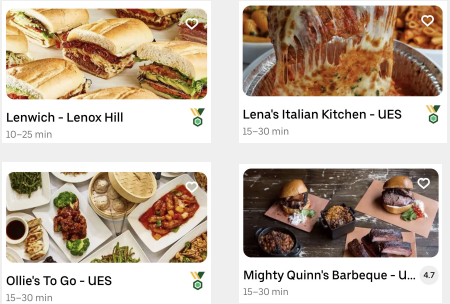
Delivery Check

User



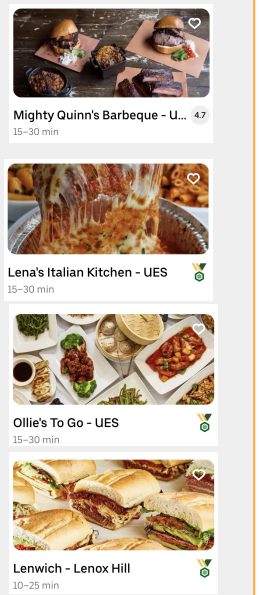
- Order history
- Price sensitivity
- Cuisine preference

Deliverable Merchants



RecSys Model

Ranked Merchants



Marketing Personalizer | User Location Determination

Estimate probability user will receive an eats order in a location *h*

$$\text{score}(u, h) = \hat{f}_\theta \left(n_{uh}, X_{uhs_1}, \ldots, X_{uhs_7}; \theta \right)$$

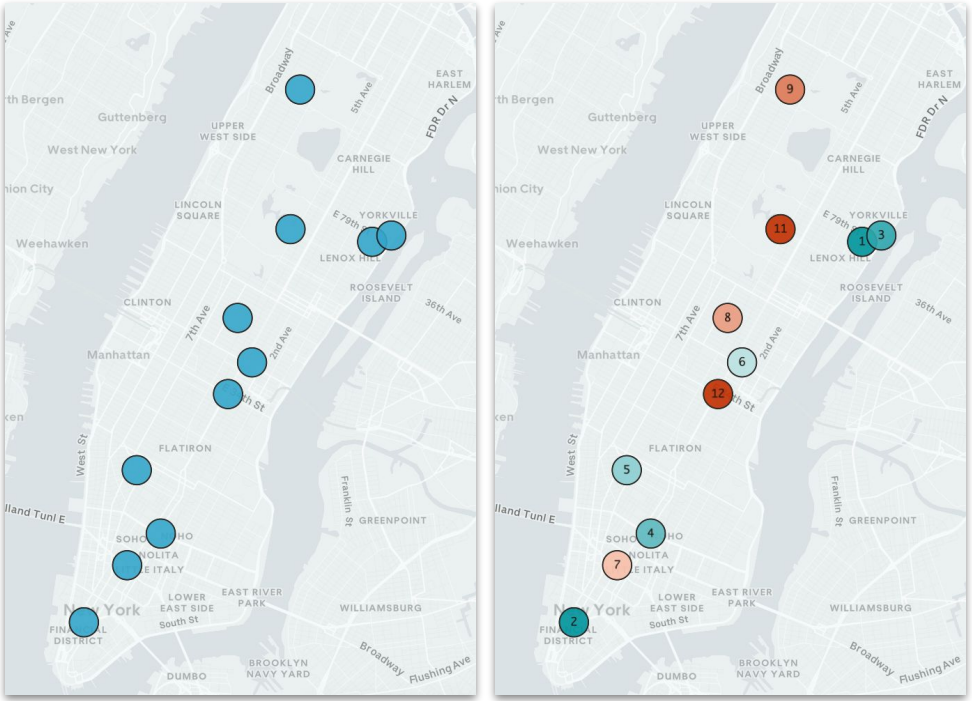
Location Ranking and Override

- Select highest scoring hex

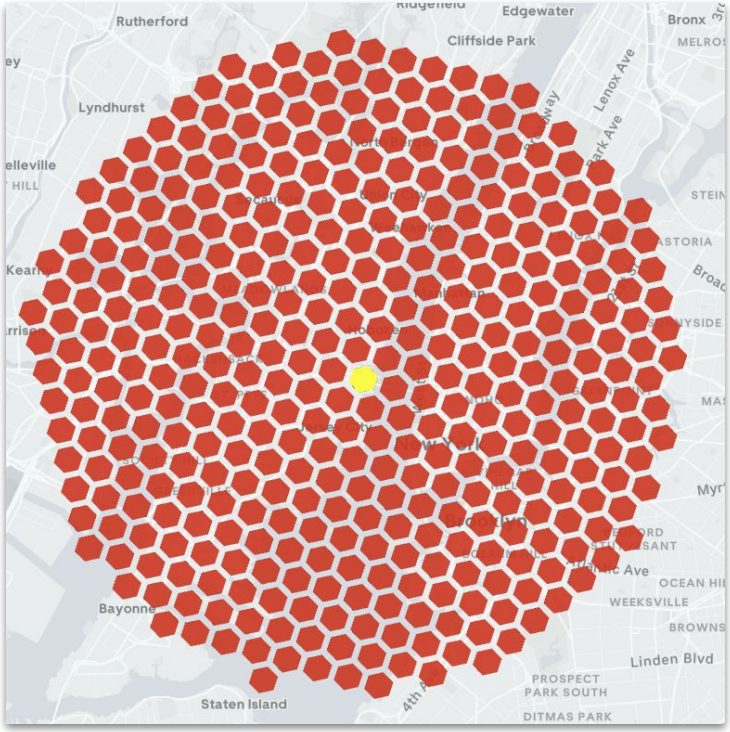
$$h_u = \underset{h \in \mathcal{H}}{\arg\max} \ \text{score}(u, h)$$

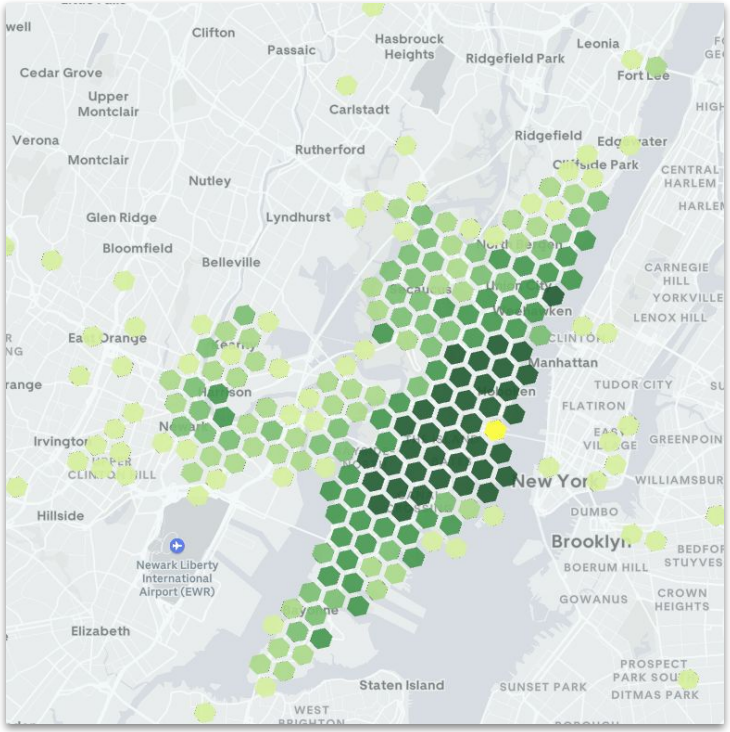
- Override the model's prediction if a user's most recent event is
  - In the last 3d
  - More than 50 miles away from h_u

- Predictions shows a 10% improvement in recall over legacy method



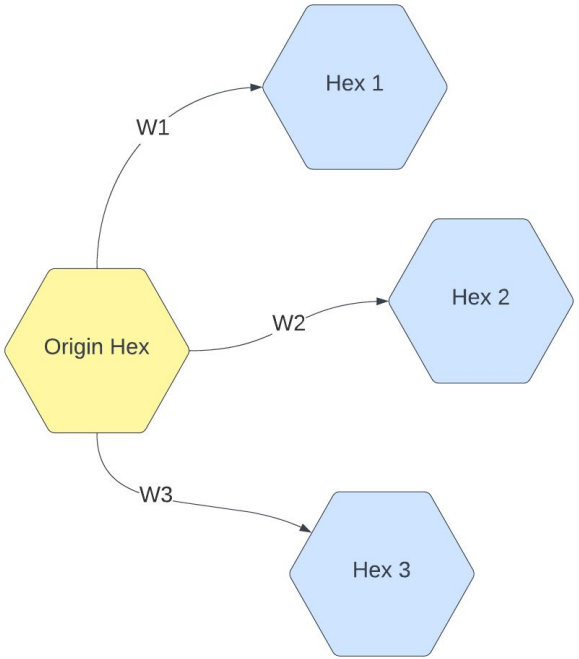
Marketing Personalizer | Merchant Delivery Hexagons





🔴 legacy delivery zones 🟡 merchant location 🟢 historical deliveries since 2020

Marketing Personalizer | Merchant Delivery Hexagons



Origin → Destination Hex Deliverability

- We know an origin → destination pair is invalid if
  - Origin hexagon has a *significant* number of orders

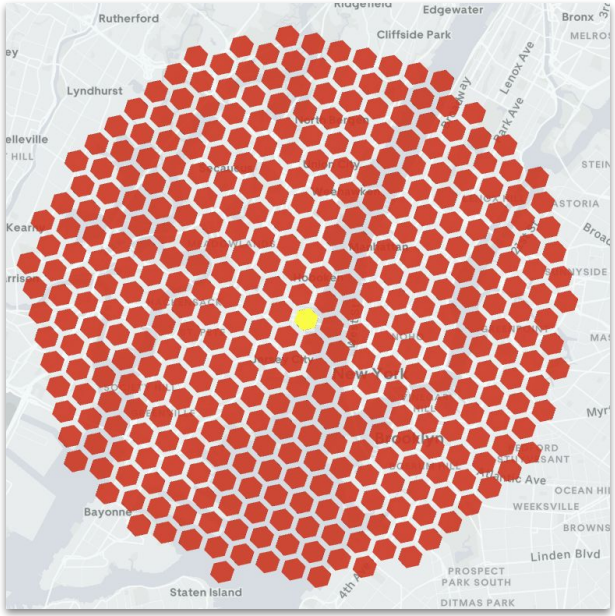
$$\sum_{i=1}^{N} W_i \geq T$$

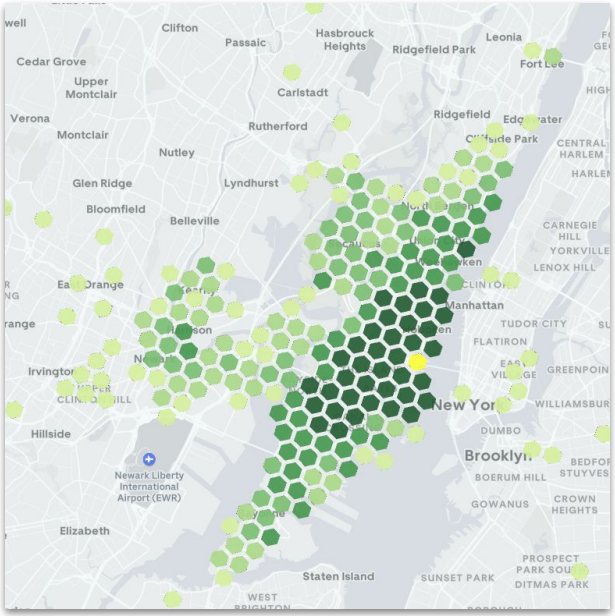
  - The number of orders to hex i is small

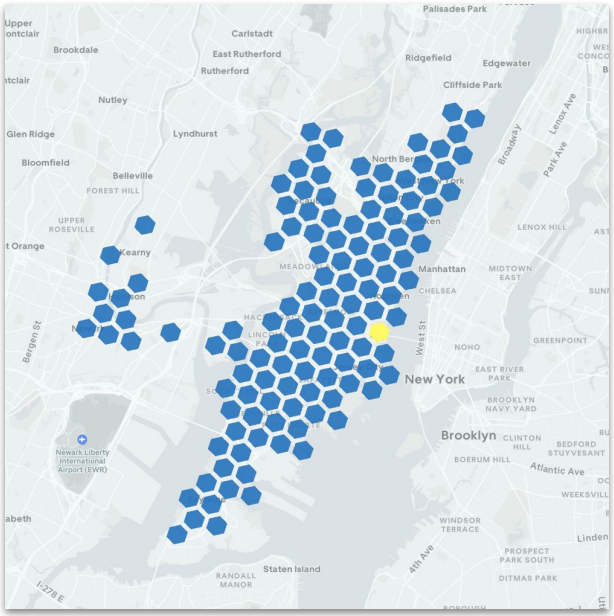
$$W_i \leq t$$

- Restrict historical delivery locations to be a *subset* of the original delivery locations

Marketing Personalizer | Merchant Delivery Hexagons



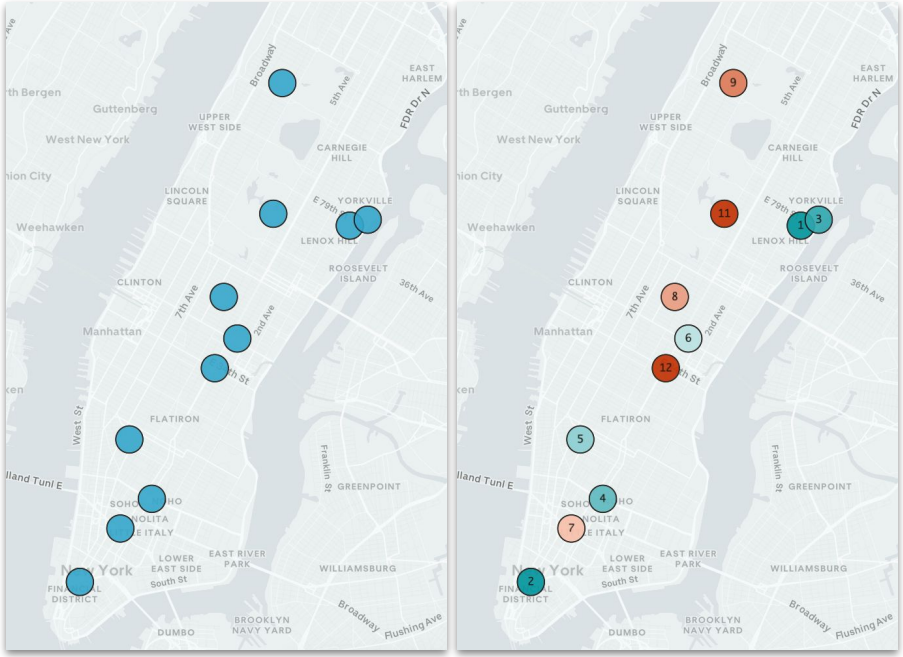




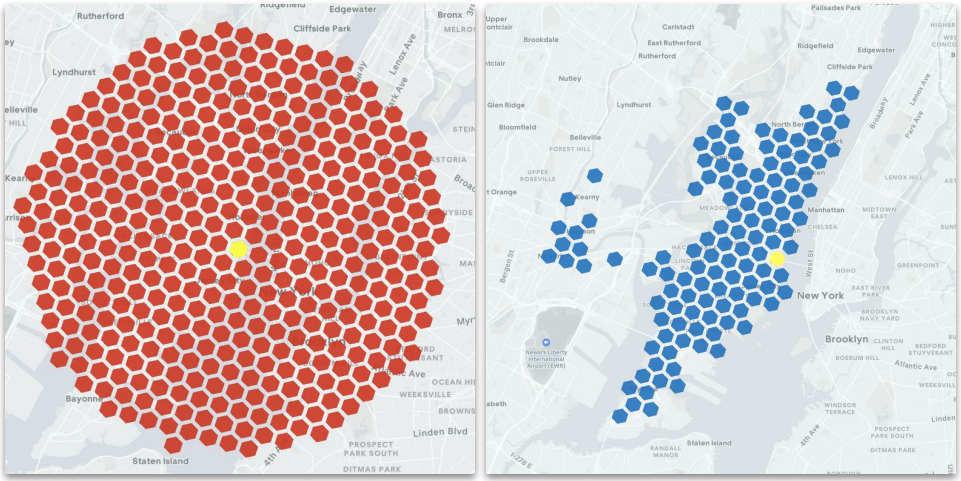
🟡 merchant location 🔴 legacy delivery zones 🟢 historical deliveries since 2020 🔵 new delivery zones

Marketing Personalizer | Offline Merchant Deliverability Checks

1. Find user delivery location

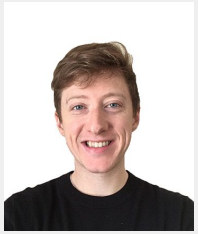


2. Identify deliverable merchants to user



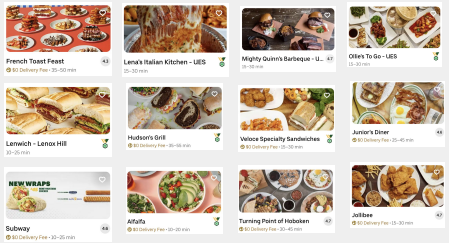
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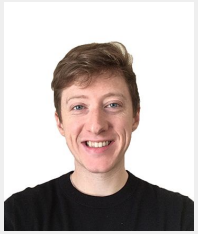
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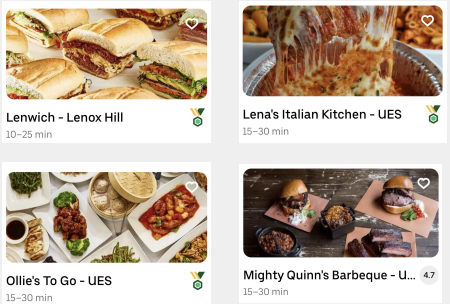
P13N Delivery Check

User



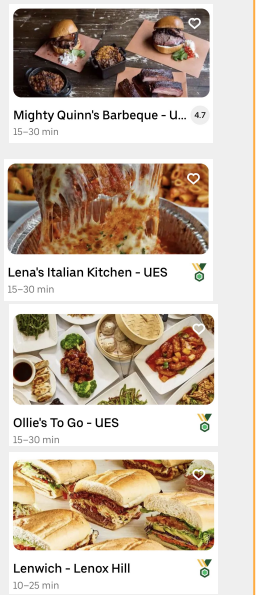
- Order history
- Price sensitivity
- Cuisine preference

Deliverable Merchants



P13N Rec Sys Model

Ranked Merchants



Question:

When recommending restaurants, what question do you ask?

Question:

When recommending restaurants, what question do you ask?

Answer:

What type of food do you like?

Question:

When recommending restaurants, what question do you ask?

Answer:

What ~~type of food~~ cuisine do you like?

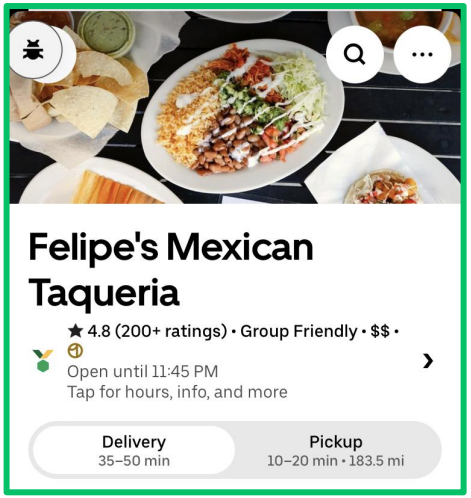
Marketing Personalizer | Cuisine

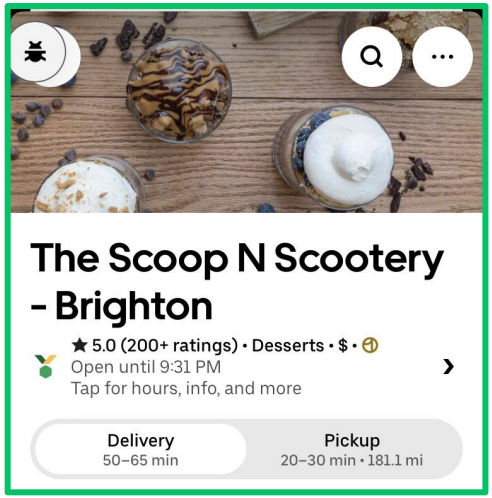
There are over **200** unique cuisine types at Uber.

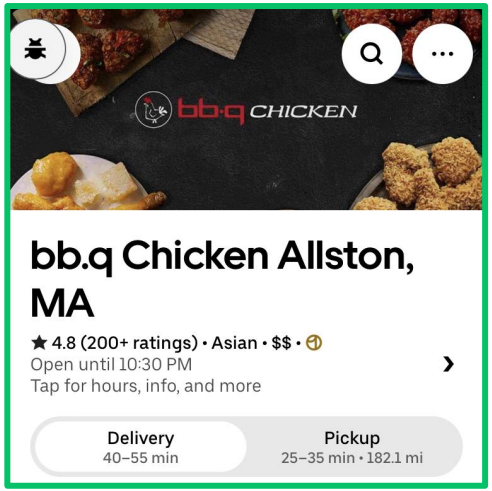
Vegetarian	Wrap	Noodle	Vegan	Korean	Himalayan	Mediterranean	Thali
Burrito	Jewish	Bulgarian	Portuguese	Irish	Swedish	Arabian	Ecuatoriana
Taco	Argentinian	Vegan	Ukrainian	Austrian	Western	Afghan	American
Wagashi	Pakistani	Portuguese	Healthy	Donut	Dumpling	Rolls	Pastry
Singaporean	DonMono	Ukrainian	Dosa	Yoshoku	Chinese	Southern	BubbleTea
Parantha	Nepalese	Healthy	Crepes	SomTum	Cupcake	Momos	Cheesesteak
Laotian	Burmese	Dosa	Dominican	Curry	Premium	Falafel	Cafe
Panamena	Bangladeshi	Crepes	Nachos	Teishoku	Ramen	Polish	Japan Curry
Comfort Food	Yucatecan	Dominican	Biryani	Deli	British	Wings	French
Pollo	Yakitori	Nachos	Hungarian	Ethiopian	E. European	Australian	Konamono
Moroccan	Romanian	Biryani	South African	Mexican	Cantonese	Jamaican	Venezuelan
Cuban	Milkshakes	Hungarian	Tempura	Syrian	Breakfast	Tom Yum	Nigerian
Burger	SouthIndian	Noodle	Vegan	Pub	African	Ice Cream	Thali
Lebanese	Kebab	Bulgarian	Portuguese	Middle Eastern	Fish & Chips	Mediterranean	Ecuatoriana

Marketing Personalizer | Cuisine

There are over **200** unique cuisine types at Uber.



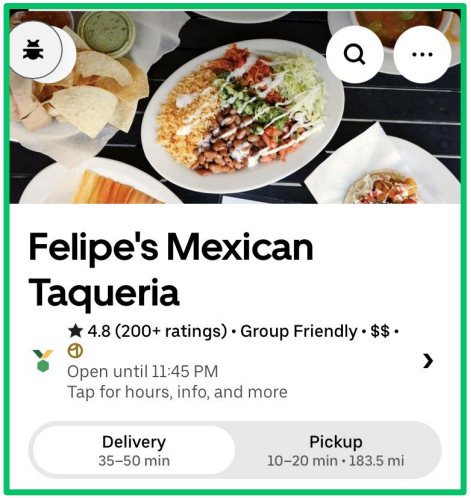


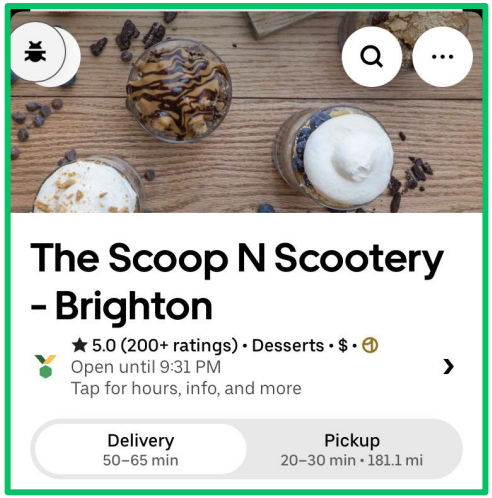


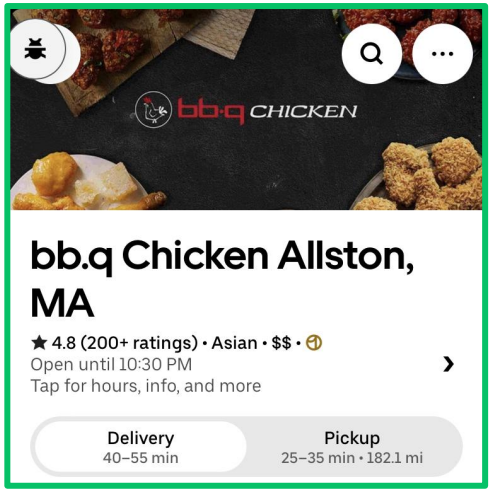
Cuisine Tags	1. Mexican 2. Burrito 3. Taco	1. Ice Cream 2. Dessert 3. Cupcake	1. Korean 2. Wings 3. Pollo

Marketing Personalizer | Cuisine

There are over **200** unique cuisine types at Uber.







Cuisine Tags	1. Mexican ~~2. Burrito~~ ~~3. Taco~~	1. Ice Cream ~~2. Dessert~~ ~~3. Cupcake~~	1. Korean ~~2. Wings~~ ~~3. Pollo~~

Marketing Personalizer | Cuisine

Cuisine	# Orders	Cuisine Affinity
Mexican	8	50%
Burrito	0	0%
Taco	0	0%
Ice Cream	4	25%
Dessert	3	19%
Cupcake	1	6%
Korean	0	0%
Wings	0	0%
Pollo	0	0%

Marketing Personalizer | Cuisine

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Cupcake	1	6%
Korean	0	0%
Wings	0	0%
Pollo	0	0%

User has a favorite Mexican restaurant they order from frequently.

Marketing Personalizer | Cuisine

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Taco	0	0%
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Cupcake	1	6%
Korean	0	0%
Wings	0	0%
Pollo	0	0%

User orders from a variety of dessert shops.

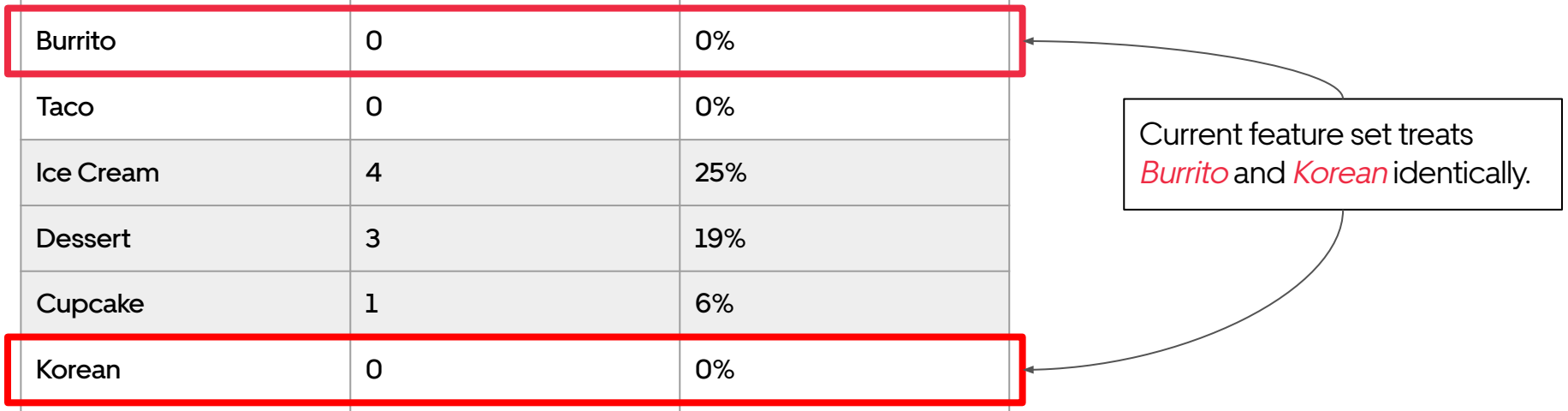
Marketing Personalizer | Cuisine

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Wings	0	0%
Pollo	0	0%

User has no engagement with Korean or Fried Chicken restaurants.

Marketing Personalizer | Cuisine

Cuisine	# Orders	Cuisine Affinity
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Taco	0	0%
Ice Cream	4	25%
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Marketing Personalizer | Cuisine

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Wings	0	0%
Pollo	0	0%

Current feature set treats *Taco* and *Wings* identically.

Marketing Personalizer | Cuisine

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Cupcake	1	6%
Korean	0	0%
Wings	0	0%
Pollo	0	0%

Takeaway:
If a user orders from a cuisine type, they likely enjoy **similar cuisines**.

Marketing Personalizer | Cuisine Similarity Embeddings

Create a co-occurrence matrix **A** such that A_{ij} = # restaurants serving **cuisine_i** and **cuisine_j**

Example

Restaurant 1	
Restaurant 2	
Restaurant 3	
Restaurant 4	

Marketing Personalizer | Cuisine Similarity Embeddings

Create a co-occurrence matrix **A** such that A_{ij} = # restaurants serving **cuisine_i** and **cuisine_j**

Example

Restaurant 1	
Restaurant 2	
Restaurant 3	
Restaurant 4	

						
	2	1	2	0	1	0
	1	1	1	0	0	0
	2	1	2	0	1	0
	0	0	0	1	0	1
	1	0	1	0	1	0
	0	0	0	1	0	2

Marketing Personalizer | Cuisine Similarity Embeddings

Normalize **A** to create such **M** such that $M_{ij} = A_{ij} / (A_{ii} * A_{jj})^{1/2}$

Example

Restaurant 1	
Restaurant 2	
Restaurant 3	
Restaurant 4	

	1	0.71	1	0	0.71	0
	0.71	1	0.71	0	0	0
	1	0.71	1	0	0.71	0
	0	0	0	1	0	0.71
	0.71	0	0.71	0	1	0
	0	0	0	0.71	0	1

Marketing Personalizer | Cuisine Similarity Embeddings

Perform matrix factorization $M = UDU^T$ and truncate low signal eigenvalues $S = U [D]_d U^T$

Example

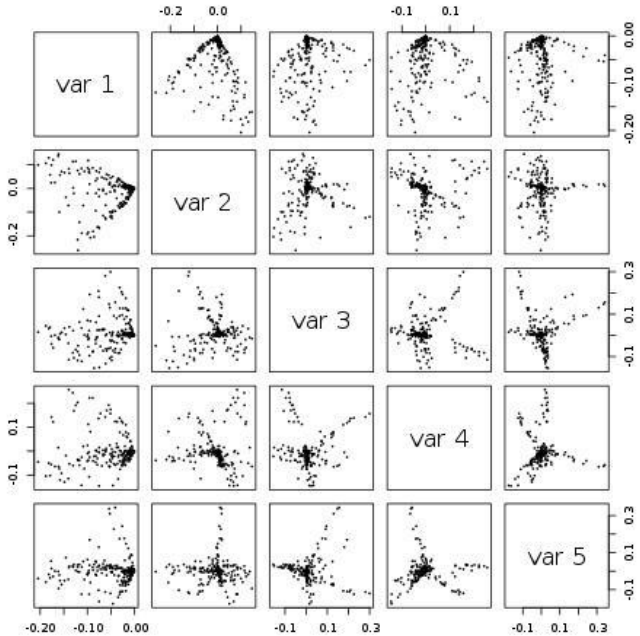
Restaurant 1	
Restaurant 2	
Restaurant 3	
Restaurant 4	

	1	0.71	1	0	0.71	0
	0.71	1	0.71	0	0.5	0
	1	0.71	1	0	0.71	0
	0	0	0	1	0	0.85
	0.71	0.5	0.71	0	1	0
	0	0	0	0.85	0	1

Marketing Personalizer | Cuisine Similarity Embeddings

Perform matrix factorization **M** = **UDU**^{T} and apply a **varimax** rotation to leading eigenvectors [**U**]$_{d}$

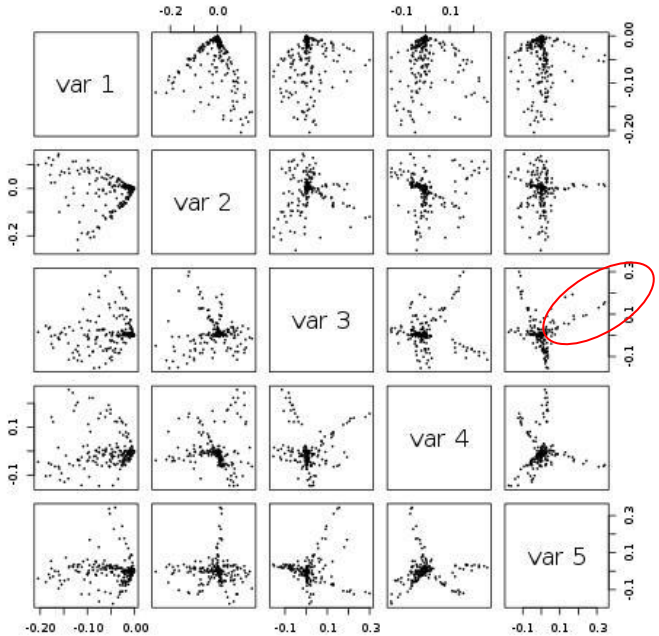
Unrotated Eigenvectors [**U**]$_{d}$



Marketing Personalizer | Cuisine Similarity Embeddings

Perform matrix factorization **M** = **UDU**^{T} and apply a **varimax** rotation to leading eigenvectors [**U**]$_{d}$

Unrotated Eigenvectors [**U**]$_{d}$

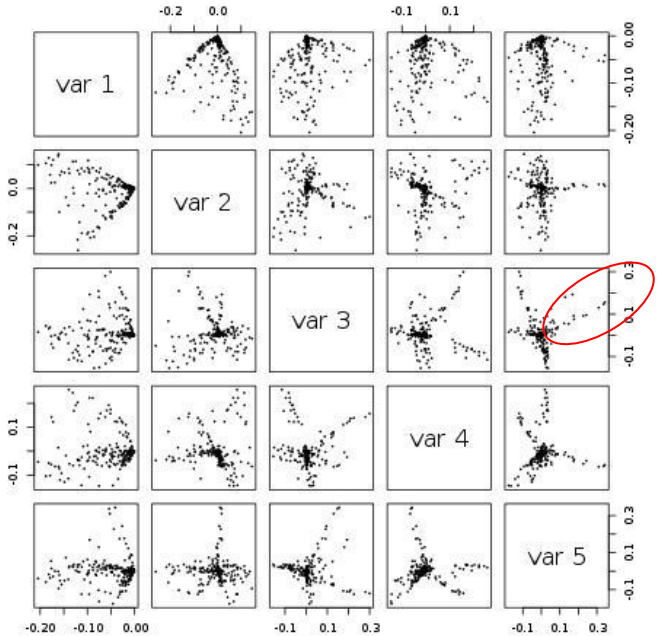


Radial Streaks

Marketing Personalizer | Cuisine Similarity Embeddings

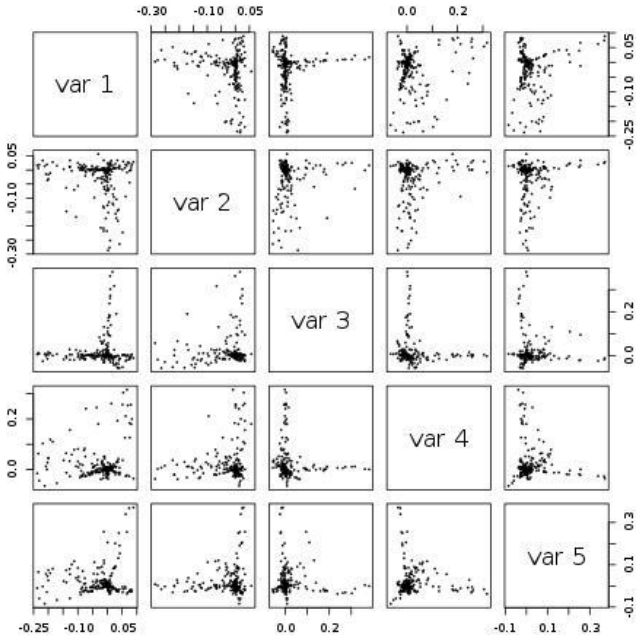
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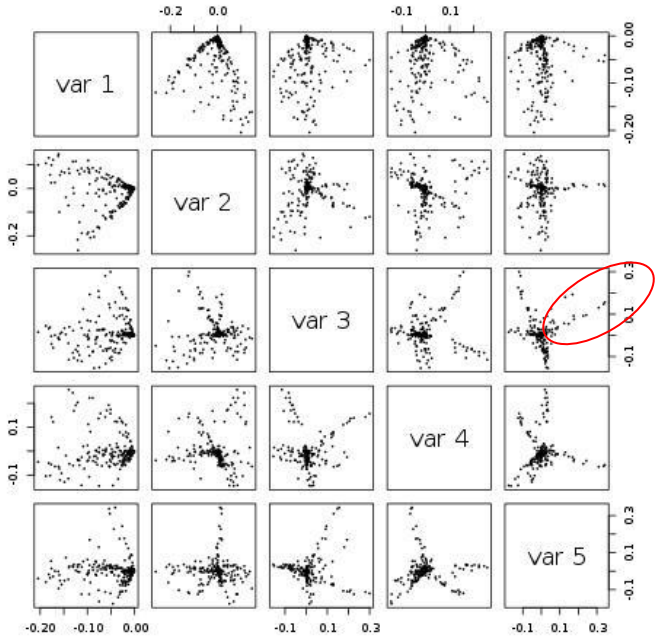
Rotated Eigenvectors varimax ([**U**]$_{d}$)



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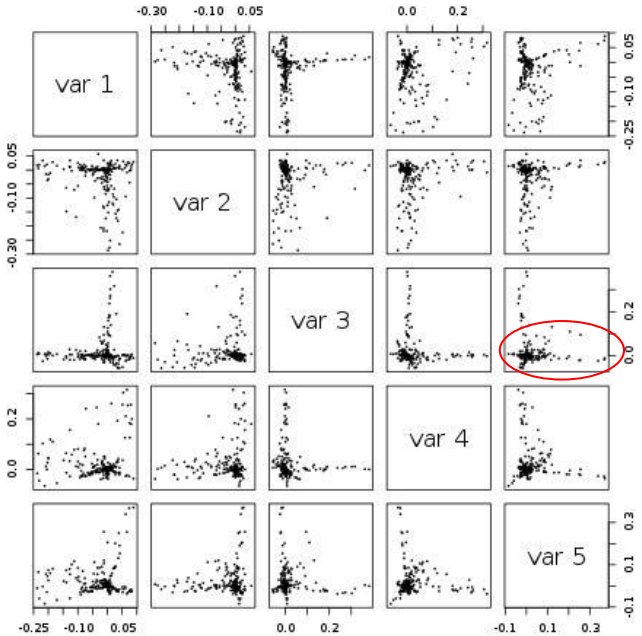
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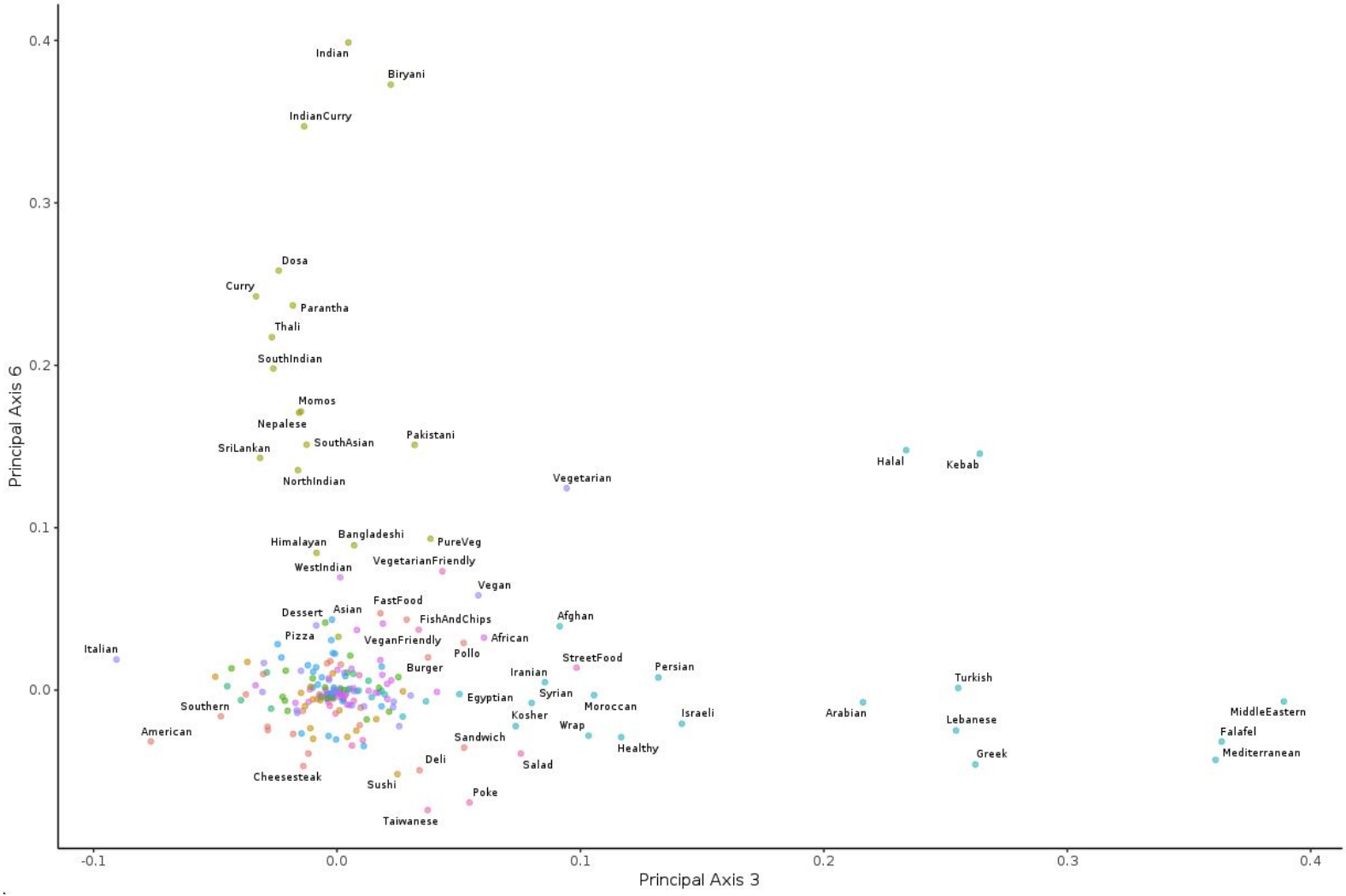


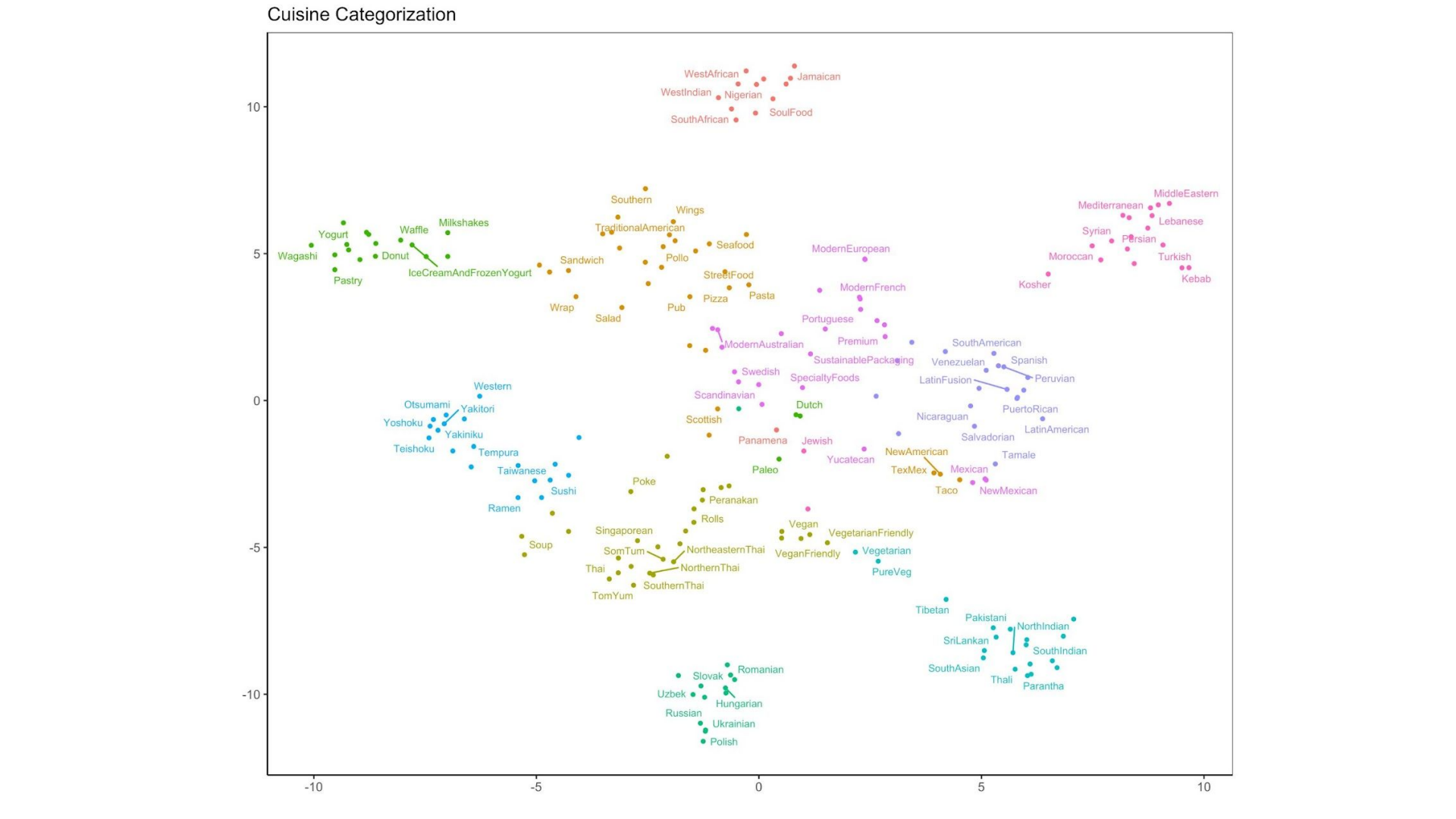
Radial Streaks

Rotated Eigenvectors varimax ([**U**]$_{d}$)



Axis-aligned Streaks


Principal Axis 6
Principal Axis 3
0.4
0.3
0.2
0.1
0.0
-0.1
0.0
0.1
0.2
0.3
0.4
Indian
Biryani
IndianCurry
Dosa
Curry
Parantha
Thali
SouthIndian
Momos
Nepalese
SriLankan
SouthAsian
Pakistani
NorthIndian
Vegetarian
Halal
Kebab
Himalayan
Bangladeshi
PureVeg
WestIndian
VegetarianFriendly
Vegan
Dessert
Asian
FastFood
FishAndChips
Afghan
Pizza
VeganFriendly
African
Italian
Pollo
StreetFood
Burger
Iranian
Persian
Turkish
Egyptian
Syrian
Southern
Moroccan
Kosher
Israeli
Arabian
MiddleEastern
Wrap
Lebanese
American
Sandwich
Falafel
Healthy
Greek
Deli
Salad
Mediterranean
Cheesesteak
Sushi
Poke
Taiwanese


Cuisine Categorization
WestAfrican
Jamaican
WestIndian
Nigerian
SoulFood
SouthAfrican
Southern
Wings
TraditionalAmerican
Seafood
Pollo
Sandwich
StreetFood
Pizza
Pasta
Wrap
Salad
Pub
Yogurt
Waffle
Milkshakes
Wagashi
Donut
IceCreamAndFrozenYogurt
Pastry
ModernEuropean
ModernFrench
Portuguese
Premium
ModernAustralian
SustainablePackaging
Swedish
SpecialtyFoods
Scandinavian
Dutch
Scottish
Panamena
Jewish
Yucatecan
Paleo
MiddleEastern
Mediterranean
Lebanese
Syrian
Persian
Moroccan
Turkish
Kebab
Kosher
SouthAmerican
Venezuelan
Spanish
Peruvian
LatinFusion
PuertoRican
Nicaraguan
LatinAmerican
Salvadorian
NewAmerican
Tamale
TexMex
Mexican
Taco
NewMexican
Western
Otsumami
Yakitori
Yoshoku
Yakiniku
Teishoku
Tempura
Taiwanese
Sushi
Ramen
Soup
Poke
Peranakan
Rolls
Singaporean
SomTum
NortheasternThai
Thai
NorthernThai
SouthernThai
TomYum
Vegan
VegetarianFriendly
VeganFriendly
Vegetarian
PureVeg
Tibetan
Pakistani
NorthIndian
SriLankan
SouthIndian
SouthAsian
Thali
Parantha
Romanian
Slovak
Uzbek
Hungarian
Russian
Ukrainian
Polish
10
5
0
-5
-10
-10
-5
0
5
10

Marketing Personalizer | Cuisine Similarity Embeddings

Cuisine	Meta-Cuisine	# Orders	Cuisine Affinity	Meta-Cuisine Affinity
Mexican	Mexican	8	50%	53%
Burrito	Mexican	0	0%	53%
Taco	Mexican	0	0%	53%
Ice Cream	Dessert	4	25%	47%
Dessert	Dessert	3	19%	47%
Cupcake	Dessert	1	6%	47%
Korean	Asian	0	0%	0%
Wings	American	0	0%	0%
Pollo	American	0	0%	0%

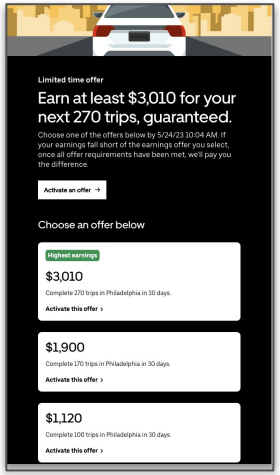
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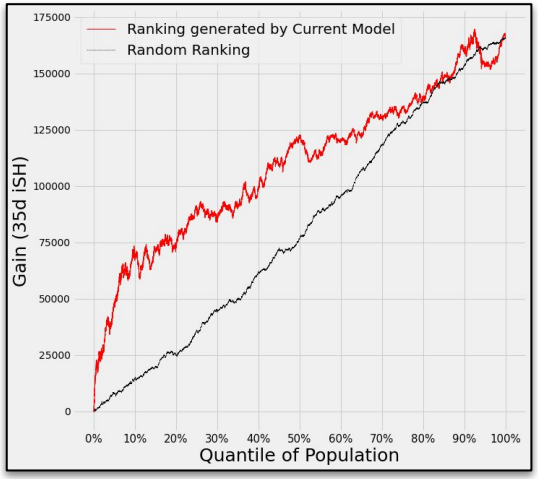
Cuisine	Meta-Cuisine	# Orders	Smoothed Cuisine Affinity
Mexican	Mexican	8	51%
Burrito	Mexican	0	21%
Taco	Mexican	0	21%
Ice Cream	Dessert	4	33%
Dessert	Dessert	3	30%
Cupcake	Dessert	1	22%
Korean	Asian	0	0%
Wings	American	0	0%
Pollo	American	0	0%

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Growing Uber's Core Businesses





Personalization

I work on personalizing consumer's (eaters, riders, grocery and retail users, ...) experiences in out-of-app channels to drive product awareness, adoption, and growth. This oftentimes manifests in **recommending restaurants, merchants, and offers** to consumers to encourage product exploration.

Targeting

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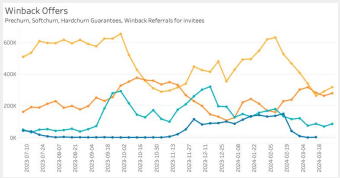
Marketing Targeting | Earner Incentives

Earner



- Trip History
- Acceptance Rate
- Vehicle Type

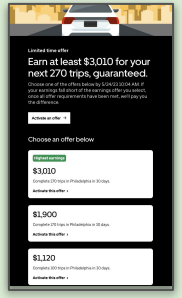
Marketplace Dynamics



- Budget
- Supply conditions

Uplift Targeting Model

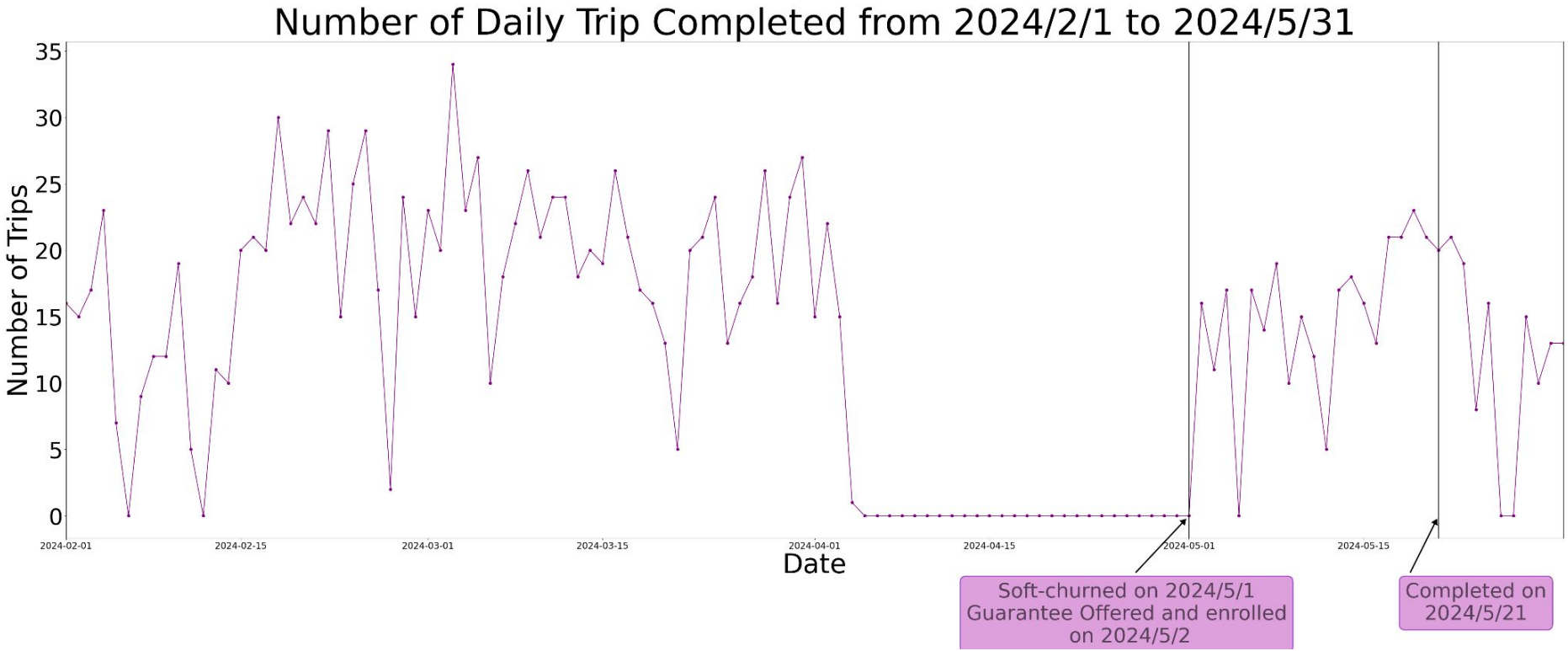
Targeted



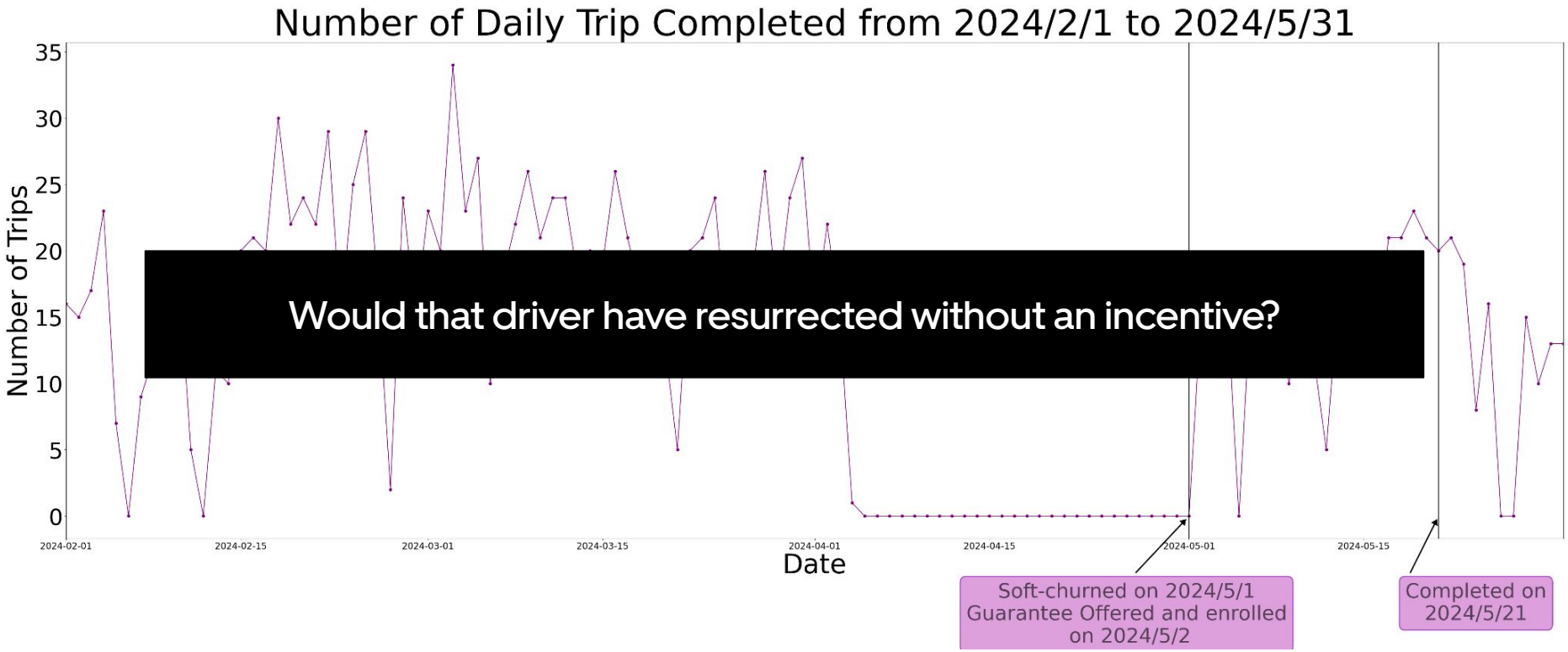
Not-Targeted

No incentive

Marketing Targeting | Earner Incentives



Marketing Targeting | Earner Incentives



Marketing Targeting | Meta Learners

Ideally, we only give drivers an incentive if they provided *incremental* supply to the marketplace.

Targeting Approach

- Easy to estimate ATE via holdout that randomly assigns treatment

$$\hat{\tau} = \hat{\mu_T} - \hat{\mu_C}$$

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- Not useful for targeting individual earners. Instead we need the *individual treatment effect* (ITE)

$$\hat{\tau}_i = \hat{\mu_T}(i) - \hat{\mu_C}(i)$$

- Unobservable counterfactual - fundamental problem of causal inference

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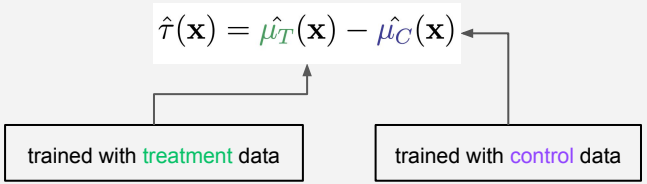
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Meta Learners

- Can we *predict* what the user *would have done* if assigned to the other treatment group?

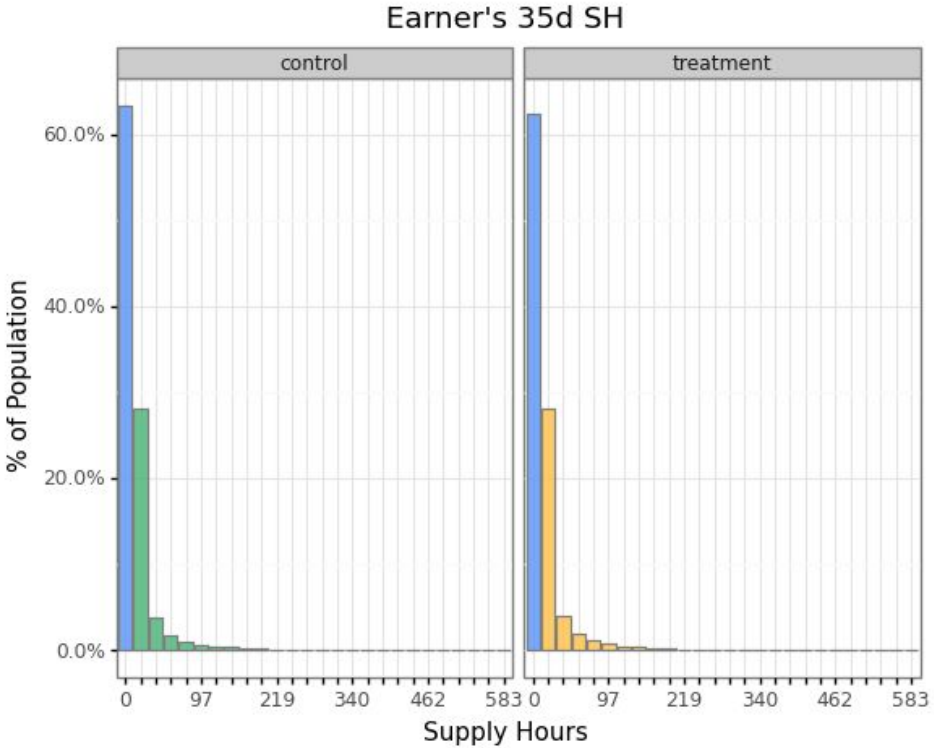


- Then we can target based on the estimated ITE

$$\psi(\mathbf{x}) = \begin{cases} \text{Targeted} & \hat{\tau}(\mathbf{x}) > 0 \\ \text{Not Targeted} & \hat{\tau}(\mathbf{x}) \leq 0 \end{cases}$$

Marketing Targeting | Meta Learners

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trained with treatment data

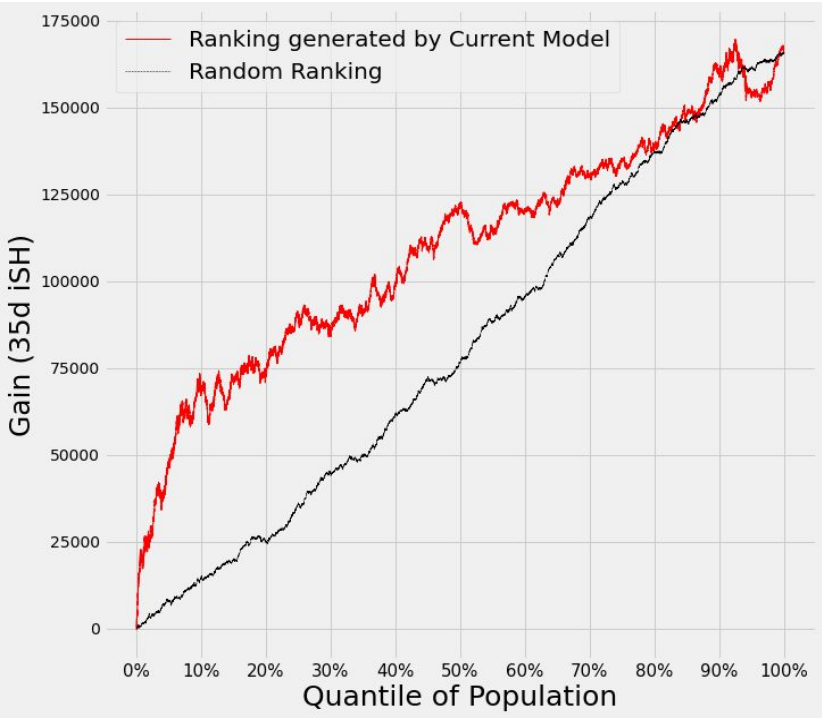
trained with control data

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Marketing Targeting | Budget Constraints

We introduce budget constraints to the targeting model to limit the cost to the business and improve efficiency of the incentive lever.

Constrain # Offers

$$\underset{z_i \in \{0,1\}}{\text{maximize}} \quad \sum_{i=1}^{N} z_i \hat{\tau}(x_i)$$

$$\text{subject to} \quad \sum_{i=1}^{N} z_i \leq M$$

- Pros: Simple
- Cons: no natural choice for M

Constrain Budget

$$\underset{z_i \in \{0,1\}}{\text{maximize}} \quad \sum_{i=1}^{N} z_i \hat{\tau}(x_i)$$

$$\text{subject to} \quad \sum_{i=1}^{N} z_i \hat{c}(x_i) \leq B$$

- Pros: Budget constraints match business function
- Cons: No cost projection

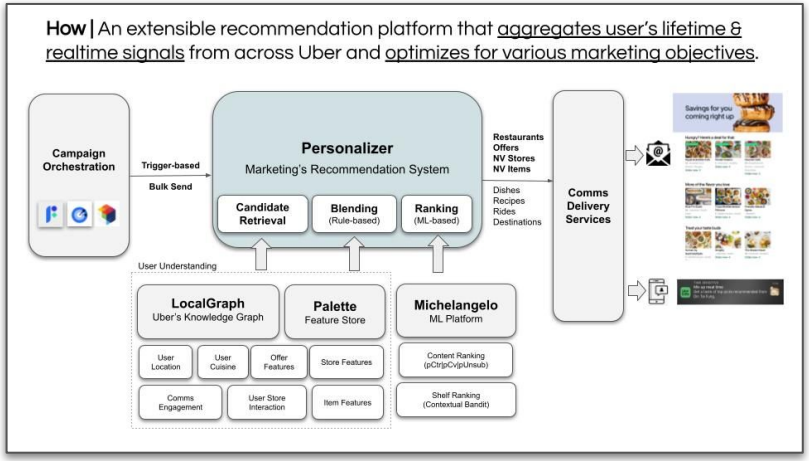
Constrain Efficiency

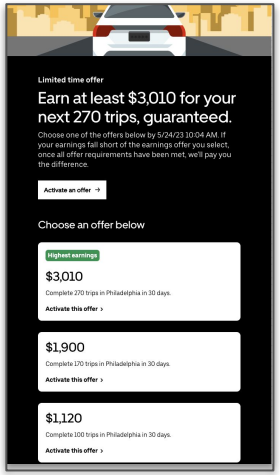
$$\underset{z_i \in \{0,1\}}{\text{maximize}} \quad \sum_{i=1}^{N} z_i \hat{\tau}(x_i)$$

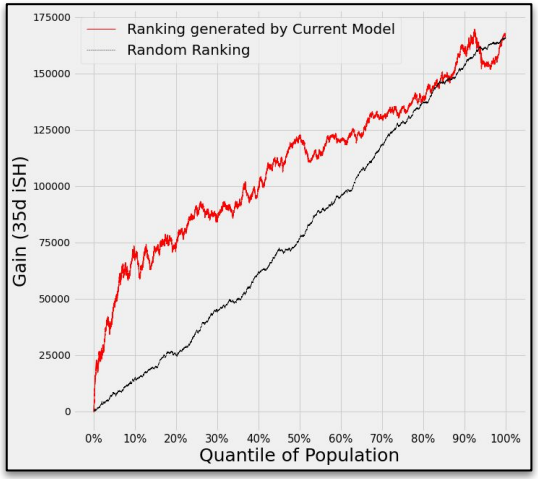
$$\text{subject to} \quad \frac{\sum_{i=1}^{N} z_i \hat{c}(x_i)}{\sum_{i=1}^{N} z_i \hat{\tau}(x_i)} \leq T$$

- Pros: Efficiency constraint allows Uber to reallocate to more efficient levers
- Cons: No cost projection

Growing Uber's Core Businesses







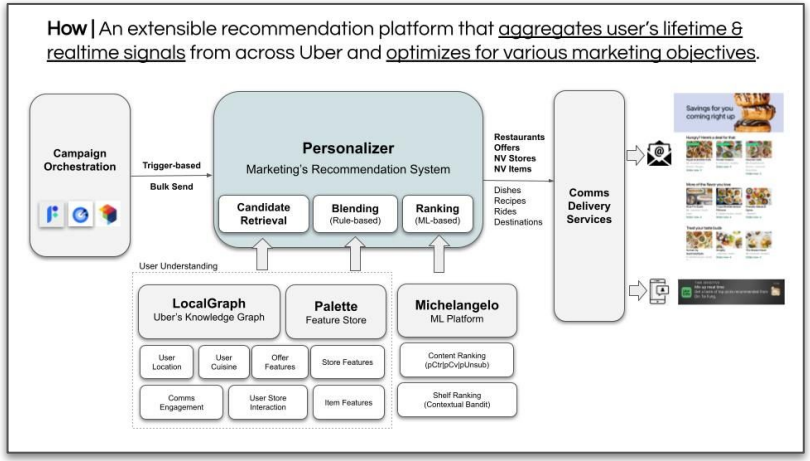
Personalization

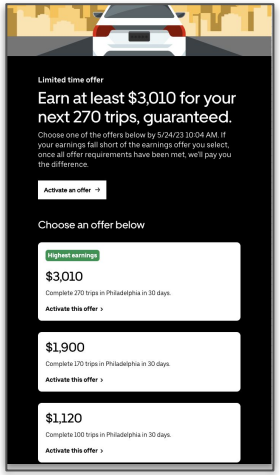
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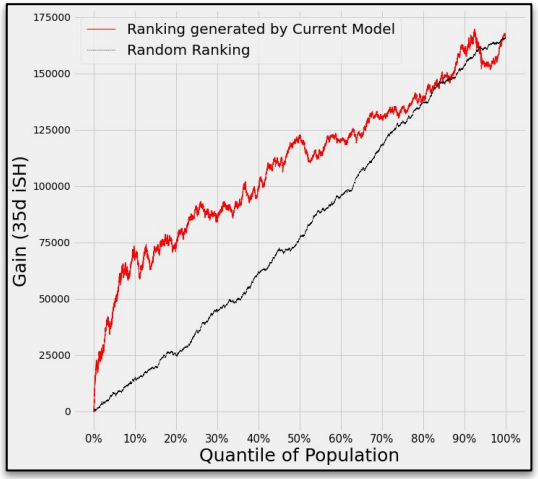
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Growing Uber's Core Businesses







Personalization

- **ML:** Rec Models (CF, MF, LTR, Two-tower embeddings)
- **Tech Stack:**
  - Day to Day: *SQL, Python*
  - Feature Eng: *PySpark, Airflow*
  - Model Training: *Ray training on Docker*
  - Serving: *Go Service API*

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- **ML:** Classical ML Models, Meta Learners, Causal Forrests
- **Tech Stack:**
  - Day to Day: *SQL, Python*
  - Feature Eng: *PySpark, Airflow*
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Discussion

Uber